MATH 152 **QUIZ 9** NAME _____

Mrs. Bonny Tighe 25 points

11.11,10.5 TA_____ Wed 4/24/2024

1. Find the Taylor series for centered at $a=3$
$f(x)=\ln(1+2x)$

2. Find the Taylor polynomial $T_3(x)$ for the function $f(x)=\sin x \ \ at \ \ a=\pi/6$ and use it to estimate $\sin(29\pi/180)$

3. Use the **TAYLOR'S INEQUALITY** given to approximate the function $f(x)=\frac{1}{(20+x)^{2/3}}$ by a Taylor polynomial of degree 3 at $a = 7$

How accurate is this approximation when $8<x<9$ If $\left|f^{(n+1)}(x)\right|\leq M$ for $|x-a|\leq d$, then the remainder $R_n(x)\leq\frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a|\leq d$

4. a) Find the vertex, focus and directrix. Sketch the graph

$y^2-2y+x+3=0$

b) Find the center, foci, and vertices. Sketch the graph

$x^2+4x+4y^2-8y+4=0$

c) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$16y^2-9x^2+64y+-36x-116=0$

5. Find the equation for the conic that satisfies the given conditions.

a) Parabola with vertex (2, -1) and directrix of $x = 3$

b) Ellipse with foci at (- 1, -1) and (7, - 1) and a vertex at (-2, -1)

c) Hyperbola with center (2, 3) a vertex at (2, 0) and a focus at (2, -1)

MATH 152 **QUIZ 9A** NAME _____

Mrs. Bonny Tighe 25 points

11.11, 10.5 TA_____Wed

4/24/2024

1. Use a known series to approximate $e^{0.2}$ to three nonzero terms

2. Find the Taylor polynomial $T_3(x)$ for the function $f(x) = \cos x \ \ at \ \ a = \pi/6$ and use it to estimate $\cos(29\pi/180)$

4. Use the **TAYLOR'S INEQUALITY** given below to approximate the function $f(x) = \sqrt{x+2}$ by a Taylor polynomial of degree 3 at x = 2.
How accurate is this approximation when $-1 < x < 5$? If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

5. Sketch the following parabola. State the vertex, focus and directrix.
$y^2 + 2y - 4x = 11$

6. Find equations for the following:

Parabola: vertex (1,0) directrix y = - 2

Ellipse: foci (0, 2), (6, 2) vertex (8, 2)

Hyperbola: vertices (1, 3), (-5, 3) focus (2, 3)

7. Find the center, foci, and vertices. Sketch the graph

$16x^2 - 32x + 9y^2 - 36y = 92$

8. Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(y-1)^2 - 9(x+2)^2 = 36$